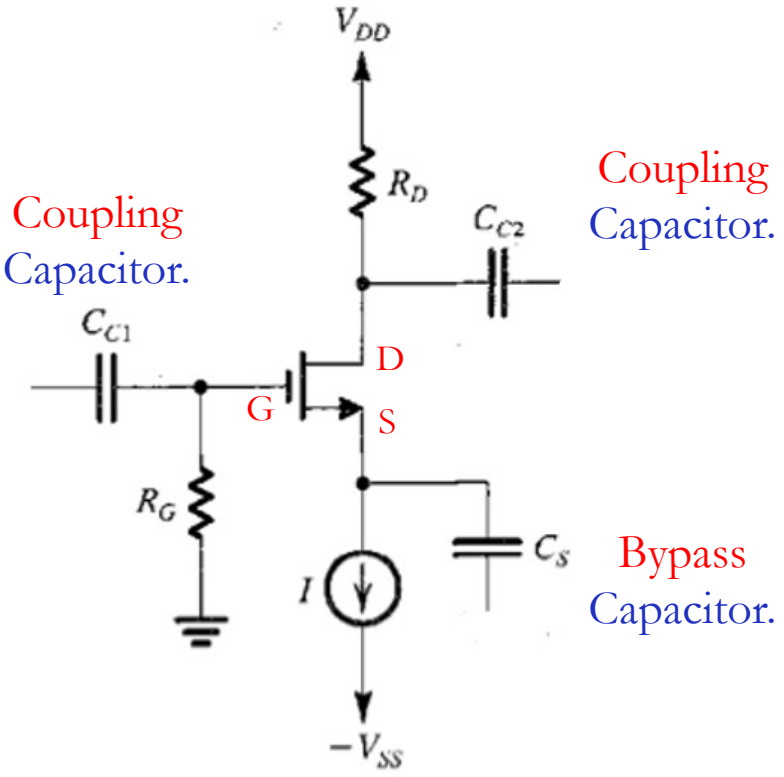
Field Effect Transistors

- MOSFET Basic Amplifier Configurations

MOSFET Basic Amplifier Configurations

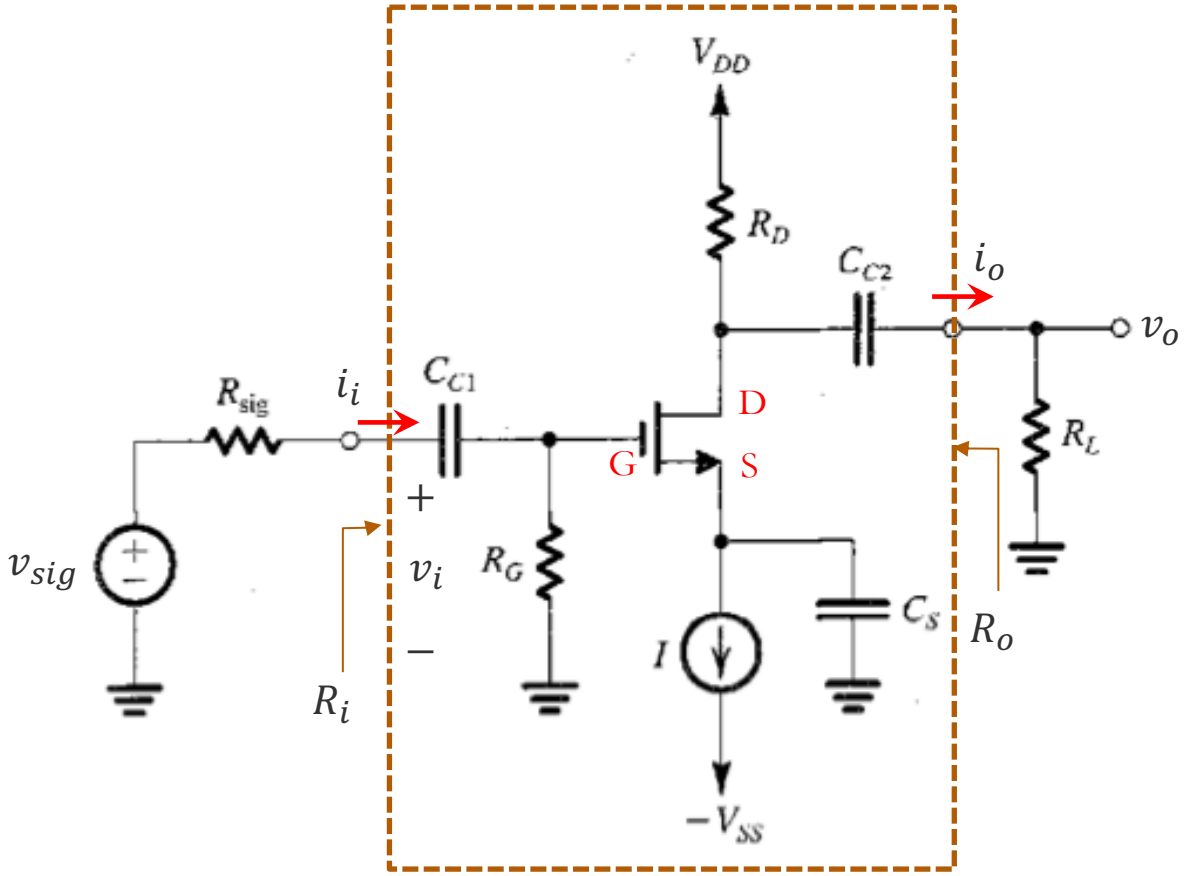
Different combinations of input-output connections are possible; each gives specific amplifier properties.

Input	Output	Ground	Name
Gate	Drain	Source	Common-source (CS)
Gate	Source	Drain	Common-drain (CD)
Source	Drain	Gate	Common-gate (CG)



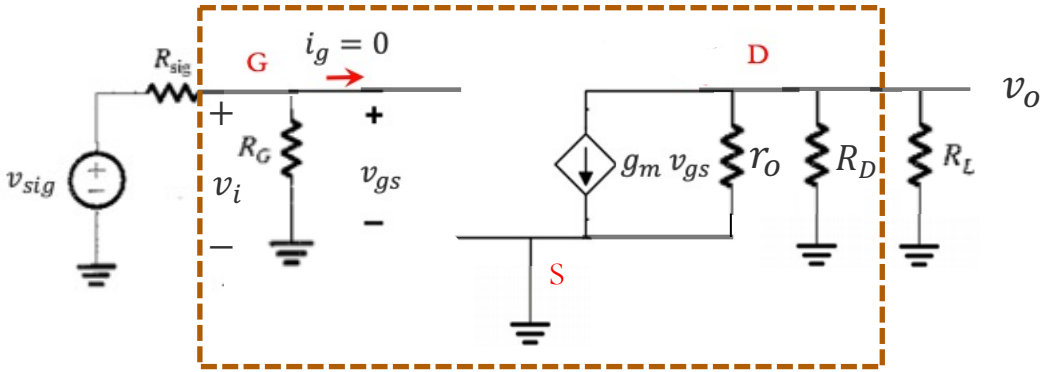
The Common-source (CS) Amplifier

This configuration works as "all-purpose amplifier": high voltage and current gains.



The Common-source (CS) Amplifier

This configuration works as "all-purpose amplifier": high voltage and current gains.



$$v_i = v_{gs} \qquad v_i = v_{sig}\frac{R_G}{R_{sig}+R_G}$$

$$v_o = -g_m v_{gs}\,(r_o \parallel R_D \parallel R_L)$$

$$v_o = -g_m v_i\,(r_o \parallel R_D \parallel R_L)$$

$$A_v = \frac{v_o}{v_i} = -g_m\,(r_o \parallel R_D \parallel R_L)$$

Voltage gain

$$A_v = \frac{v_o}{v_{sig}} = -g_m \frac{R_G}{R_{sig}+R_G}\,(r_o \parallel R_D \parallel R_L)$$

Overall Voltage gain

Open-circuit Voltage gain (no load)

$$A_{vo} = \frac{v_o}{v_i} = -g_m\,(r_o \parallel R_D)$$

Input Resistance

$$R_i = R_G$$

Output Resistance
short-circuit source v_{sig}

$$R_o = r_o \parallel R_D$$

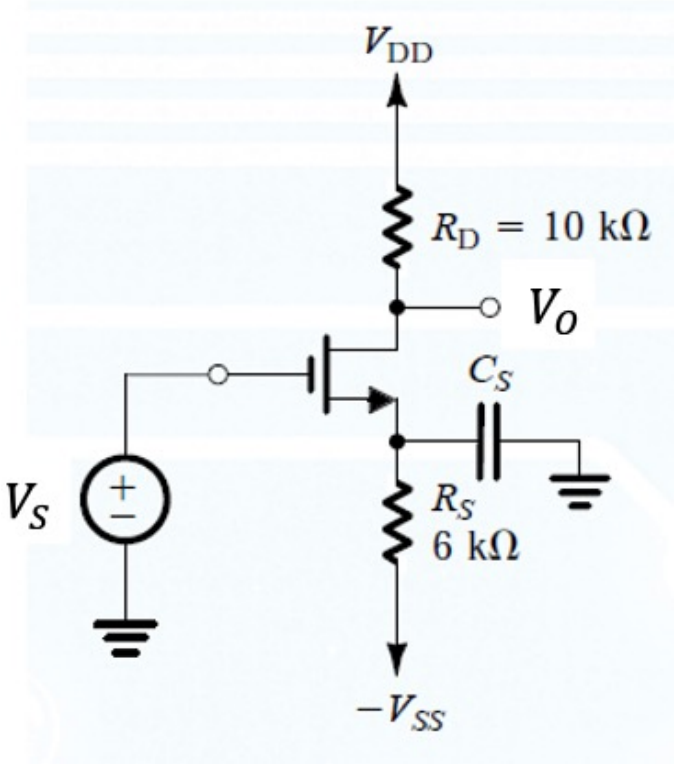
Current gain

$$A_i = \frac{i_o}{i_i} = \frac{v_o}{R_L}\cdot\frac{R_i}{v_i}$$

$$A_i = \frac{i_o}{i_i} = A_v \cdot \frac{R_i}{R_L}$$

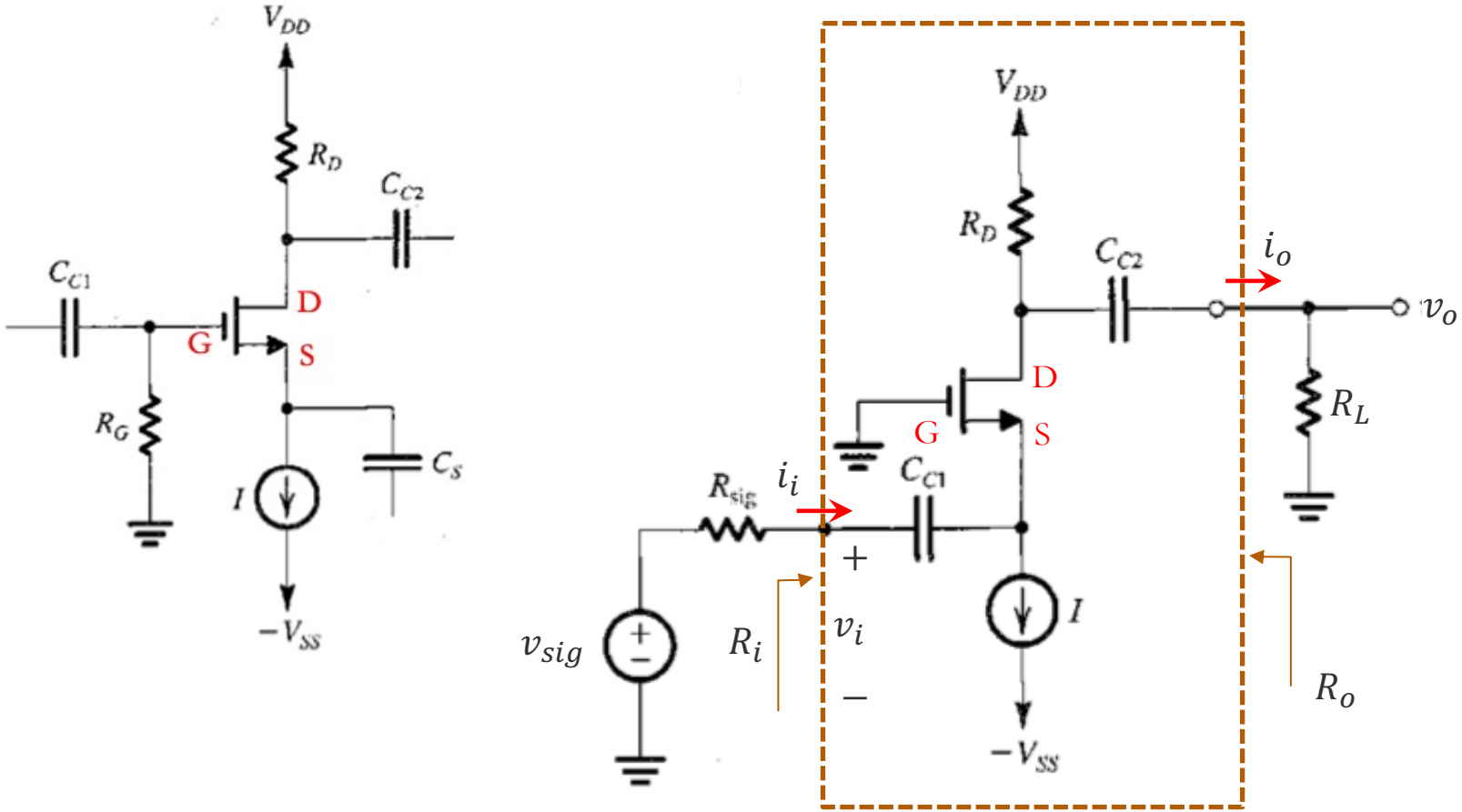
Example

Analyze the amplifier given below.



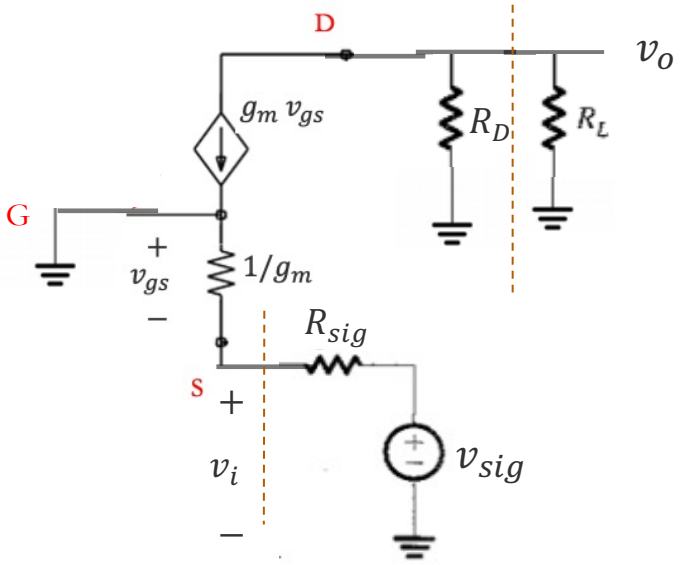
The Common-gate (CG) Amplifier

This configuration works as a "current buffer": unity current gain.



The Common-gate (CG) Amplifier

This configuration works as a "current buffer": unity current gain.



$$v_i = -v_{gs}$$

$$g_m v_{gs} = -\frac{v_o}{R_D \parallel R_L} \qquad g_m v_i = \frac{v_o}{R_D \parallel R_L}$$

$$A_v = \frac{v_o}{v_i} = g_m (R_D \parallel R_L)$$

Input Resistance

$$R_i = \frac{1}{g_m}$$

Output Resistance
short-circuit source v_{sig}

$$R_o = R_D$$

Current gain

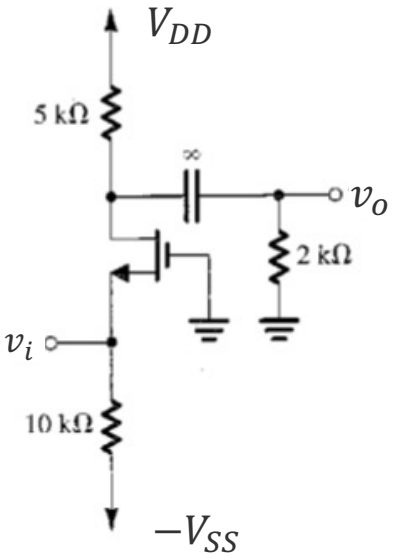
$$A_i = \frac{i_o}{i_i} = A_v \cdot \frac{R_i}{R_L}$$

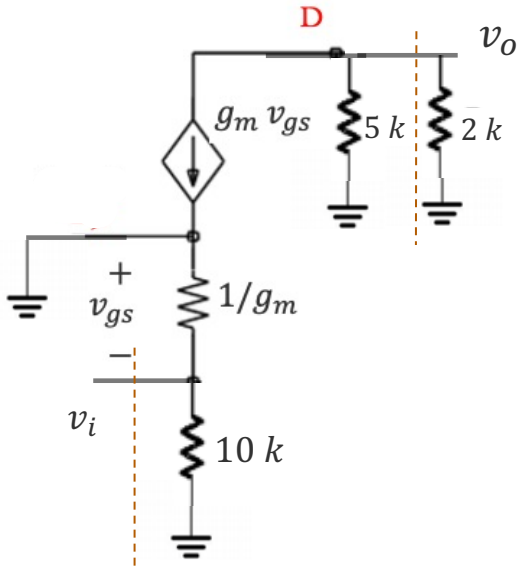
$$A_i = g_m (R_D \parallel R_L) \cdot \frac{1/g_m}{R_L}$$

$$A_i = \frac{(R_D \parallel R_L)}{R_L} \leq 1$$

Example

Analyze the amplifier given below. Given g_m=5 mA/V and r_o very high.





$$v_i = -v_{gs}$$

$$v_o = -g_m v_{gs}\,(5\ k \parallel 2\ k)$$

$$v_o = g_m v_i\,(5\ k \parallel 2\ k)$$

$$A_v = \frac{v_o}{v_i} = g_m(5\ k \parallel 2\ k)$$

$$A_v = 7.14\ V/V$$

Input Resistance

$$R_i = (10\ k \parallel 1/g_m)$$

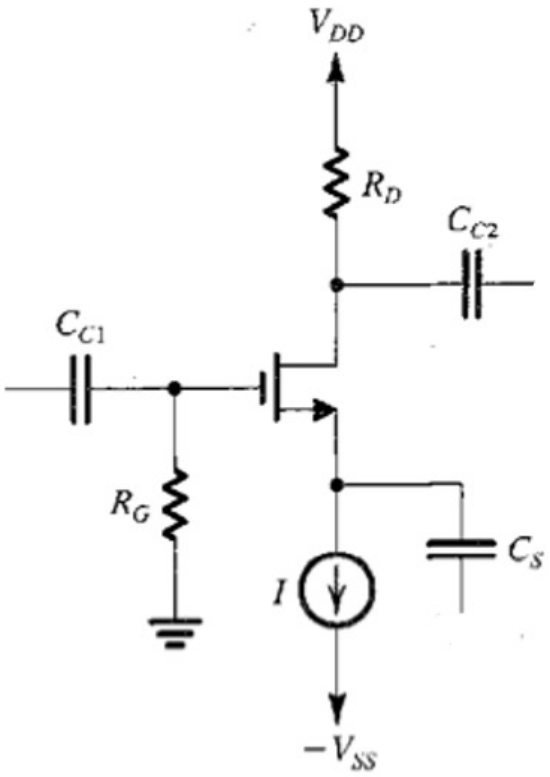
$$R_i = 196.1\ \Omega$$

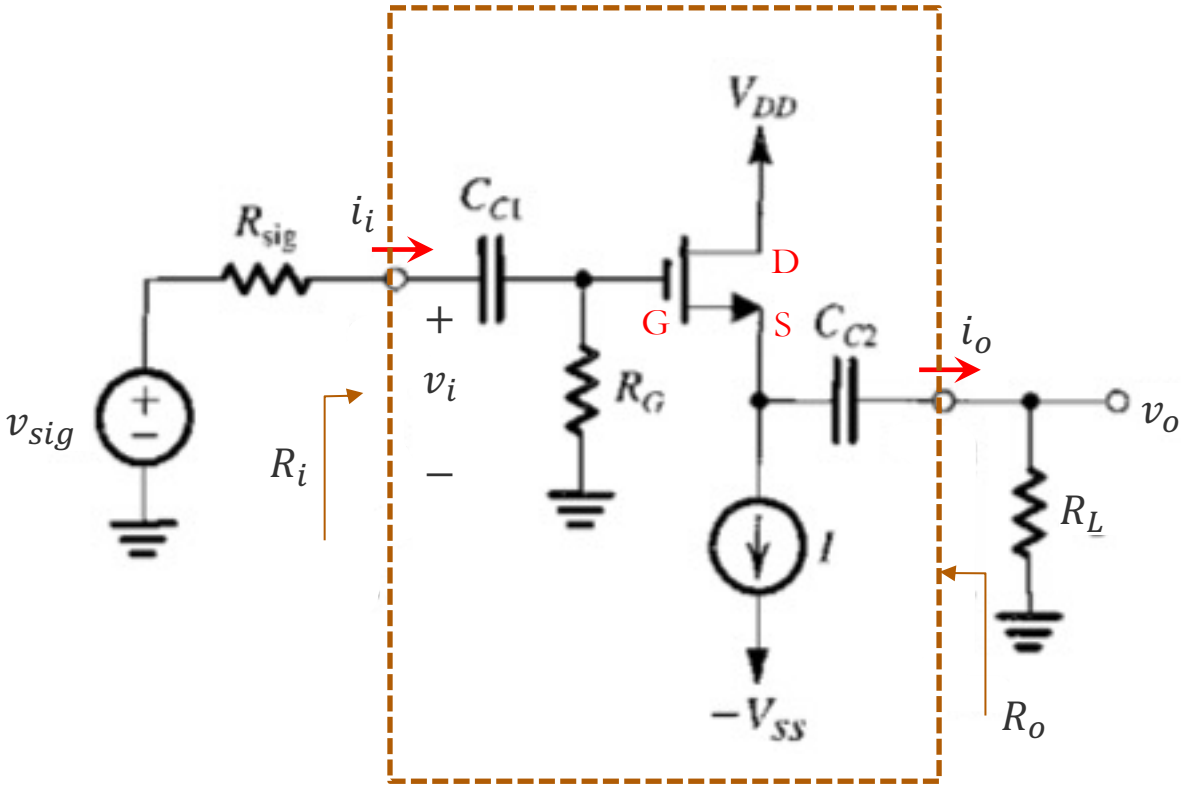
Output Resistance
short-circuit source v_i

$$R_o = 5\ k\Omega$$

The Common-drain (CD) Amplifier

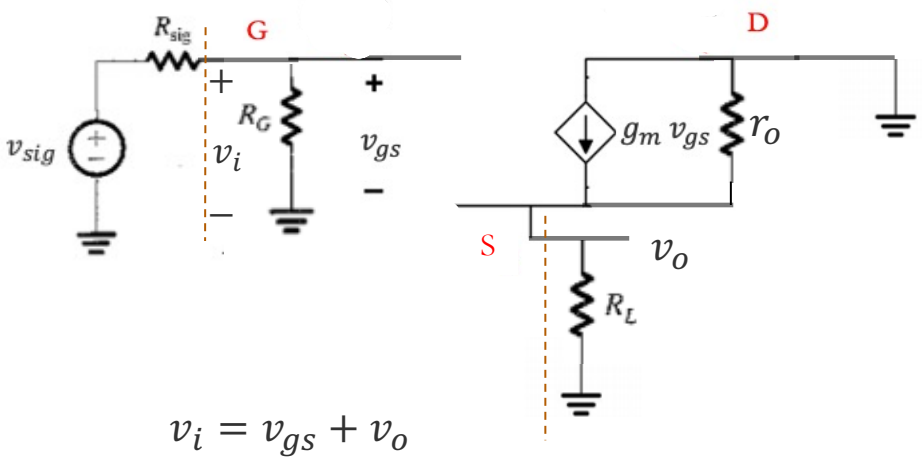
This configuration works as a "source-follower": unity voltage gain.





The Common-drain (CD) Amplifier

This configuration works as a "source-follower": unity voltage gain.



$$v_i = v_{gs} + v_o$$

$$g_m v_{gs} + \frac{0 - v_o}{r_o} = \frac{v_o}{R_L}$$

$$g_m(v_i - v_o) + \frac{0 - v_o}{r_o} = \frac{v_o}{R_L}$$

$$A_v = \frac{v_o}{v_i} = \frac{g_m}{g_m + 1/R_L + 1/r_o} \leq 1$$

Input Resistance

$$R_i = R_G$$

Output Resistance
short-circuit source v_{sig} $\quad R_o = \frac{v_o}{i_o}$

$$i_o = g_m(-v_o) + \frac{0 - v_o}{r_o}$$

$$R_o = \frac{v_o}{i_o} = \frac{1}{g_m + 1/r_o} \approx \frac{1}{g_m}$$

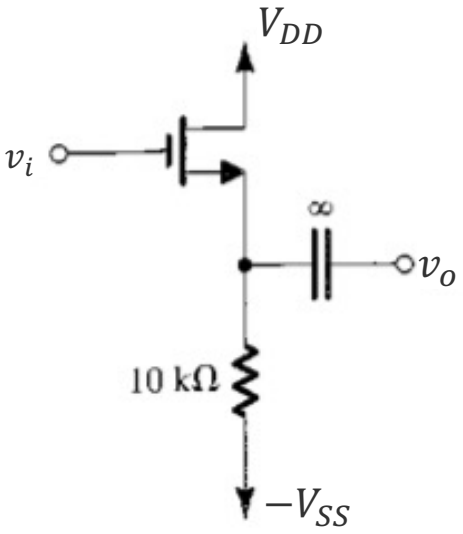
Current gain

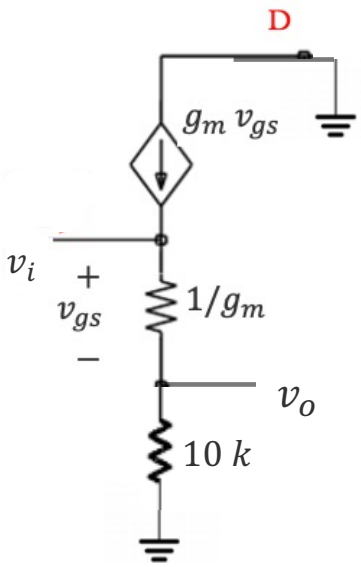
$$A_i = \frac{i_o}{i_i} = A_v \cdot \frac{R_i}{R_L}$$

$$A_i = \frac{g_m}{g_m + 1/R_L + 1/r_o} \frac{R_G}{R_L}$$

Examples

1) Analyze the amplifier given below. Given g_m=5 mA/V and r_o very high.





$$v_i = v_{gs} + v_o$$

$$v_o = g_m v_{gs}\,(10\ k)$$

$$v_o = g_m(v_i - v_o)(10\ k)$$

$$A_v = \frac{v_o}{v_i} = \frac{g_m\,(10\ k)}{1 + g_m\,(10\ k)}$$

$$A_v = 0.98\ V/V$$

Input Resistance

$$R_i = \infty$$

Output Resistance
short-circuit source v_i

$$R_o = \frac{v_o}{i_o}$$

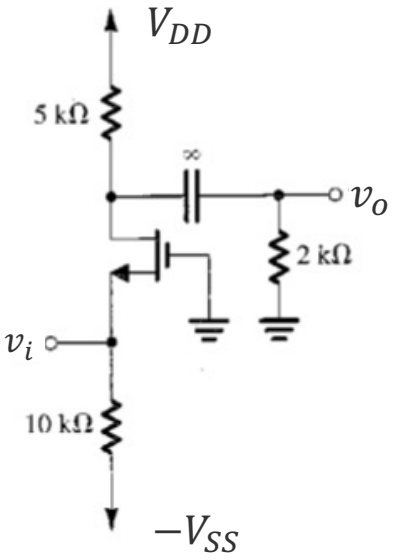
$$i_o = g_m v_{gs} - \frac{v_o}{10\ k}$$

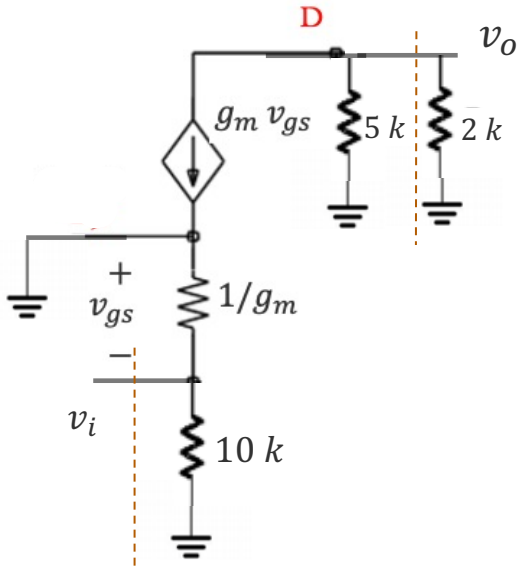
$$i_o = -g_m v_o - \frac{v_o}{10\ k}$$

$$R_o = \frac{1}{g_m + \frac{1}{10k}} = 196.1\ \Omega$$

Examples

2) Analyze the amplifier given below. Given g_m=5 mA/V and r_o very high.





$$v_i = -v_{gs}$$

$$v_o = -g_m v_{gs}\,(5\ k \parallel 2\ k)$$

$$v_o = g_m v_i\,(5\ k \parallel 2\ k)$$

$$A_v = \frac{v_o}{v_i} = g_m(5\ k \parallel 2\ k)$$

$$A_v = 7.14\ V/V$$

Input Resistance

$$R_i = (10\ k \parallel 1/g_m)$$

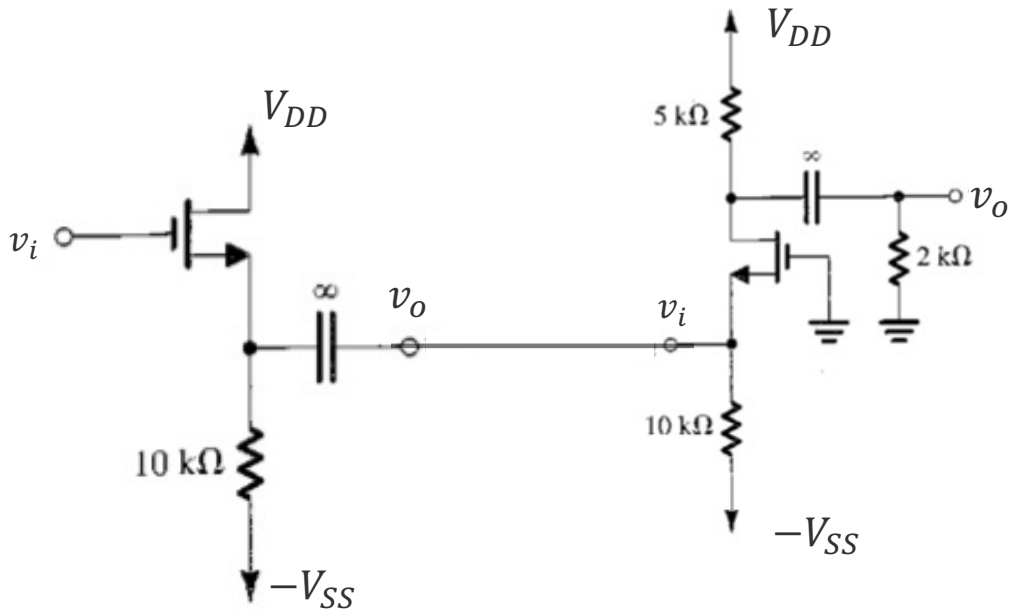
$$R_i = 196.1\ \Omega$$

Output Resistance
short-circuit source v_i

$$R_o = 5\ k\Omega$$

Examples

3) Connect the two circuits.



Multi-stage Amplifier Design

Voltage gain

$$A_v = \left(\frac{v_o}{v_i}\right)_1 \cdot \left(\frac{v_o}{v_i}\right)_2$$

$$A_v = 7\ V/V$$

Input Resistance

$$R_i = \infty$$

Output Resistance

$$R_o = 5\ k\Omega$$

Q & A

Send me an email if
you have any
questions.